Roll No.: _

Amrita Vishwa Vidyapeetham, Coimbatore Campus

B.Tech End semester Examinations – March 2026

Sixth Semester

Computer Science and Engineering

23CSE465 MOBILE APPLICATION DEVELOPMENT

SET-2

Duration: Two hours

Maximum : 50 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO01	To design Flutter applications using basic UI components.
CO02	To design Flutter applications using Advanced UI components.
CO03	To develop applications to fetch data from a remote server and advanced operations.
CO04	To develop mobile applications with DB connections.

Answer all questions

PART – A

1. Front-End Development in Flutter [15 Marks][CO02][BTL3]

Design and implement a Plant Care Companion Mobile Application using Flutter that helps users manage and monitor their plants. The application should allow users to add plants, track watering schedules, and maintain plant care records. The focus should be on creative UI design, interactive widgets, and responsive layouts.

i. Plant Gallery Screen

Create a Home Screen with an AppBar titled "Plant Care Companion" and a Floating Action Button (FAB) to add a new plant.

Display plants using a GridView gallery layout where each plant appears as a Card containing:

- Plant Image
- Plant Name
- Watering Frequency (e.g., every 3 days)

Each plant card should also display a small status indicator showing whether the plant needs watering.

ii. Add Plant Screen

Provide input widgets to add a plant, including:

- TextField for Plant Name
- Dropdown menu to select plant type (Indoor, Outdoor, Succulent, etc.)
- Image Picker to upload a plant photo from the gallery or camera
- Slider or Stepper widget to set watering frequency (in days)

Include a Save button to dynamically add the plant to the gallery.

iii. Plant Details Screen

Use GestureDetector or InkWell so that tapping a plant card opens a Plant Details Screen displaying:

- Plant Image
- Plant Name
- Plant Type

- Watering Frequency
- Last Watered Date

Include a "Mark as Watered" button that updates the plant status.

iv. Edit Plant Screen

Allow users to modify plant details such as plant name, watering schedule, or plant image.

v. Themes, Styles, and Interactivity

Apply green-themed UI styling, custom icons, and plant-related images.
Use Hero animations for smooth transitions between the plant gallery and details screen. Include an AlertDialog before deleting a plant. Demonstrate the use of Stateful and Stateless widgets along with state management techniques such as setState, Provider, BLoC, or GetX. Add a search feature to filter plants by name or type.

2. Backend Integration and Smart Features [15 Marks] [CO03, CO04] [BTL4]

Connect the Plant Care Companion App to a backend system to implement full CRUD operations. All changes must dynamically reflect in the user interface.

Backend Requirements

- Create: Add new plant records and store them in the backend database.
- Read: Retrieve plant details dynamically from the backend.
- Update: Modify plant information, such as watering schedule or plant details.
- Delete: Remove plant records with confirmation dialogs.

Students may choose one backend technology:

i. Firebase Firestore – Real-time synchronization of plant data.
ii. REST API (HTTP + JSON) – Connect the app to an external API service.
iii. Local Database (SQFLite) – Store plant data locally.
iv. MongoDB – Connect through a Node.js or Python backend.

Mandatory Smart Features

- Watering Reminder Notifications: Send notifications reminding users when a plant needs watering.
- Location Tagging: Allow users to attach the GPS location where the plant is placed (e.g., garden, balcony) and display the location in the plant details screen.

Part-B

3. Predict the Output of the Following Dart Programs[**5** Marks] [CO0**1**] [BTL**3**]

a.

```
void main() {
  var list1 = [
    [1, 2],
    [3, 4]
  ];

  var list2 = list1.toList();

  list1[0][0] = 10;

  print(list2);
}
```

b.

```
void main() {
  int? a;
  int b = 5;

  var result1 = a ?? b * 2;
  var result2 = (a ?? b) * 2;

  print(result1);
  print(result2);}
```

4. Explain the purpose of the FutureBuilder widget in Flutter when working with asynchronous data. [5 Marks] [CO01] [BTL1]
5. Differentiate between Push Notifications and Local Notifications in mobile applications. [5 Marks] [CO03] [BTL2]
6. Differentiate between Firebase Authentication and Firebase Firestore in Flutter applications. [5 Marks] [CO04] [BTL2]

CO	Marks	BTL	Marks
CO01	**10**	BTL 1	**5**
CO02	**15**	BTL 2	**10**
CO03	**12**	BTL 3	**20**
CO04	**13**	BTL 4	**15**
CO05		BTL 5	
CO06		BTL 6	

Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table